

LALR PARSER

- LALR Parser is lookahead LR parser
- LALR reduces the size of the table.
- LALR parsing is same as the CLR(1) parsing. only difference is parsing table.
- Combine the states with same set of rules but different lookahead symbols.

1. For the given grammar, construct LALR parsing table.

$S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$

Solution

Augmented grammar

$S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$

Closure (I)

$S' \rightarrow \cdot S, \$$

$S \rightarrow \cdot CC, \$$

$C \rightarrow \cdot cC, (c/d)$

$C \rightarrow \cdot d, (c/d)$

Canonical LR(1) Items

$I_0: S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot CC, \$$
 $C \rightarrow \cdot cC, (c/d)$
 $C \rightarrow \cdot d, (c/d)$

goto(I_0, S)

$I_1: S' \rightarrow S \cdot, \$$

goto(I_0, C)

$I_2: S \rightarrow C \cdot C, \$$
 $C \rightarrow \cdot cC, \$$
 $C \rightarrow \cdot d, \$$

goto(I_0, c)

$I_3: C \rightarrow c \cdot C, (c/d)$
 $C \rightarrow \cdot cC, (c/d)$
 $C \rightarrow \cdot d, (c/d)$

goto(I_0, d)

$I_4: C \rightarrow d \cdot, (c/d)$

goto(I_2, C)

$I_5: S \rightarrow CC \cdot, \$$

goto(I_2, c)

$I_6: C \rightarrow c \cdot C, \$$
 $C \rightarrow \cdot cC, \$$
 $C \rightarrow \cdot d, \$$

goto(I_2, d)

$I_7: C \rightarrow d \cdot, \$$

goto(I_3, c)

$I_8: C \rightarrow cC \cdot, (c/d)$

goto(I_3, c)

$I_9: \begin{cases} C \rightarrow c \cdot C, (c/d) \\ C \rightarrow \cdot cC, (c/d) \\ C \rightarrow \cdot d, (c/d) \end{cases}$

goto(I_3, d)

$I_{10}: C \rightarrow d \cdot, (c/d)$

goto(I_6, c)

$I_{11}: C \rightarrow cC \cdot, \$$

goto(I_6, c)

$I_{12}: \begin{cases} C \rightarrow c \cdot C, \$ \\ C \rightarrow \cdot cC, \$ \\ C \rightarrow \cdot d, \$ \end{cases}$

goto(I_6, d)

$I_{13}: C \rightarrow d \cdot, \$$

Canonical LR(1) Parsing Table.

State	Action			Goto	
	c	d	\$	S	C
I_0	S_3	S_4		1	2
I_1			Accept		
I_2	S_6	S_7			5
I_3	S_3	S_4			8
I_4	r_3	r_3			
I_5			r_1		
I_6	S_6	S_7			9
I_7			r_3		
I_8	r_2	r_2			
I_9			r_2		

1. $S \rightarrow CC$
2. $C \rightarrow cC$
3. $C \rightarrow d$

→ In the LR(1) item, check whether the core item or elements (ie LR(0)) are common and there may be different lookahead.

→ If ~~common~~ core elements are common, then combine two items and perform union operation for the lookahead.

→ Consider the items I_3 and I_6 . Both these items have the same core but they differ in lookahead.

→ We can combine them and call it as I_{36}

goto(I_0, c)

goto(I_0, c) & goto(I_5, c)

I_3 : $\left\{ \begin{array}{l} C \rightarrow c.C, (c/d) \\ C \rightarrow .cC, (c/d) \\ C \rightarrow .d, (c/d) \end{array} \right\}$

I_{36} : $\left\{ \begin{array}{l} C \rightarrow c.C, (c/d/\$) \\ C \rightarrow .cC, (c/d/\$) \\ C \rightarrow .d, (c/d/\$) \end{array} \right\}$

common

[Union operation is performed for lookahead for I_3 and I_6]

I_6 : $\left\{ \begin{array}{l} C \rightarrow c.C, \$ \\ C \rightarrow .cC, \$ \\ C \rightarrow .d, \$ \end{array} \right\}$

I_4 : $\left\{ \begin{array}{l} C \rightarrow d., (c/d) \end{array} \right\}$

I_{47} : $\left\{ \begin{array}{l} C \rightarrow d., (c/d/\$) \end{array} \right\}$

I_7 : $\left\{ \begin{array}{l} C \rightarrow d., \$ \end{array} \right\}$

I_8 : $\left\{ \begin{array}{l} C \rightarrow cC., (c/d) \end{array} \right\}$

I_{89} : $\left\{ \begin{array}{l} C \rightarrow cC., (c/d/\$) \end{array} \right\}$

I_9 : $\left\{ \begin{array}{l} C \rightarrow cC., \$ \end{array} \right\}$

LALR Parsing Table

State	ACTION			GoTo	
	c	d	\$	S	C
I_0	S ₃₆	S ₄₇		1	2
I_1			Accept		
I_2	S ₃₆	S ₄₇			5
I_{36}	S ₃₆	S ₄₇			89
I_{47}	r ₃	r ₃	r ₃		
I_5			r ₁		
I_{89}	r ₂	r ₂	r ₂		

I_3 goto(I_0, c) } $\rightarrow I_{36}$
 I_6 goto(I_0, c)

I_4 goto(I_0, d) } $\rightarrow I_{47}$
 I_7 goto(I_0, d)

I_8 goto(I_3, c) } $\rightarrow I_{89}$
 I_9 goto(I_6, c)

I_{47} : $C \rightarrow d. (c/d/\$)$
 (Reduce) r₃

I_{89} : $C \rightarrow cC. (c/d/\$)$
 Reduce r₂

LALR Parsing Input

ccdd\$

Stack
\$0c36c36
89

\$0c36c36(89)
c

\$0c36c
89

\$0c2c5
5

Stack	Input Buffer	Action
\$0	ccdd\$	Shift S36
\$0c36	ccdd\$	Shift S36
\$0c36c36	dd\$	Shift S47
\$0c36c36d47	d\$	Reduce r3 (c → d)
\$0c36c36c89	d\$	Reduce r2 (C → cC) (89, d)
\$0c36c86 \$0c36c89	d\$	Reduce r2 (C → cC) (89, d)
\$0c2	d\$	Shift S47
\$0c2d47	\$	Reduce r3 (C → d)
\$0c2c5	\$	Reduce r1 (S → cC) (5, \$)
\$0S1	\$	Accept

(0, c)
(26, c)
(36, d)
(47, \$)
(89, d)
(89, d)
(2, d)
(47, \$)
(5, \$)

2. Construct LALR parser for the given grammar

$$S \rightarrow L = R \mid R$$

$$L \rightarrow * R \mid id$$

$$R \rightarrow L$$

Solution

Refer the CLR for the augmented grammar, $closure(I)$ and canonical LR(1) Items.

→ check the Item contain same core elements and combine them.

$goto(I_0, *)$

$$I_4: \left\{ \begin{array}{l} L \rightarrow *.R, =/\$ \\ R \rightarrow .L, =/\$ \\ L \rightarrow .*R, =/\$ \\ L \rightarrow .id, =/\$ \end{array} \right\}$$

$$goto(I_0, *) \& goto(I_6, *)$$

$goto(I_6, *)$

$$I_{11}: \left\{ \begin{array}{l} L \rightarrow *.R, \$ \\ R \rightarrow .L, \$ \\ L \rightarrow .*R, \$ \\ L \rightarrow .id, \$ \end{array} \right\}$$

$$I_{411}: \left\{ \begin{array}{l} L \rightarrow *.R, =/\$ \\ R \rightarrow .L, =/\$ \\ L \rightarrow .*R, =/\$ \\ L \rightarrow .id, =/\$ \end{array} \right\}$$

$goto(I_0, id)$

$$I_5: L \rightarrow id., =/\$$$

$I_{12}: L \rightarrow id., \$$

$$goto(I_0, id) \& goto(I_6, id)$$

$$I_{512}: L \rightarrow id., =/\$$$

$goto(I_4, R)$

$$I_7: L \rightarrow *R., =/\$$$

$I_{13}: L \rightarrow *R., \$$

$$goto(I_4, R) \& goto(I_{11}, R)$$

$$I_{713}: L \rightarrow *R., =/\$$$

$goto(I_4, L)$

$$I_8: R \rightarrow L., =/\$$$

$I_{10}: R \rightarrow L., \$$

$$goto(I_4, L) \& goto(I_6, L)$$

$$I_{810}: R \rightarrow L., =/\$$$

LALR Parsing Table

State	ACTION				Go To		
	<i>id</i>	=	*	\$	S	L	R
<i>P₀</i>	S ₅₁₂		S ₄₁₁		1	2	3
<i>P₁</i>				Accept			
<i>P₂</i>		S ₆		γ_5			
<i>P₃</i>				γ_2			
<i>P₄₁₁</i>	S ₅₁₂		S ₄₁₁			810	713
<i>P₅₁₂</i>		γ_4		γ_4			
<i>P₆</i>	S ₅₁₂		S ₄₁₁			810	9
<i>P₇₁₃</i>		γ_3		γ_3			
<i>P₈₁₀</i>		γ_5		γ_5			
<i>P₉</i>				γ_1			
<i>P₁₀</i>							

1. $S \rightarrow L = R$

2. $S \rightarrow R$

3. $L \rightarrow * R$

4. $L \rightarrow id$

5. $R \rightarrow L$